Integration over a Vanishing Cycle

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Introduction

Let k be a commutative field and let f be a formal power series in two indeterminates x and y over k:

$$f = \sum a_{n,m} x^n y^m.$$

Let $I(f)$ be the formal power series in one indeterminate t

$$I(f) = \sum a_{n,n} t^n.$$

The aim is to prove the following result.

Theorem 1. *If f represents an algebraic function of x and y, and if k has characteristic $p \neq 0$, then $I(f)$ represents an algebraic function of t.*

For a formal series in N indeterminates

$$g = \sum a_n x^n,$$

one defines similarly

$$I_N(g) = \sum a_{n,\dots,n} t^n,$$

and one has the following consequence.

Corollary 1. *If g is algebraic over the field $k(x_1, \dots, x_N)$, and if k has characteristic $p > 0$, then $I_N(g)$ is algebraic over $k(t)$.*

Under the stronger hypothesis $g \in k(x_1, \dots, x_N)$ this result is due to H. Furstenberg. The first purpose here is to give a geometric explanation of Furstenberg's result.

Suppose first that $k = \mathbb{C}$ and that the formal series f is convergent. Then $I(f)$ is convergent and has the integral representation

$$I(f)(t) = \frac{1}{2\pi i} \int_{xy=t} f(x,y) \, \frac{dx \wedge dy}{dt}. \tag{1}$$

Here $\pi : \mathbb{C}^2 \to \mathbb{C}$ is $(x,y) \mapsto xy$. The expression $(dx \wedge dy)/dt$ is a relative holomorphic differential on $\mathbb{C}^2 - \{(0,0)\}$, relative to π; it is also a section of the relative dualizing sheaf at $(0,0)$, in the sense of Grothendieck duality. In the coordinate y on the fibre $xy = t$ one has

$$\frac{dx \wedge dy}{dt} = \frac{d(t/y) \wedge dy}{dt} = \frac{dy}{y},$$

up to the sign fixed by the chosen order of the variables.

If f converges in a polydisc of radius R, and if $|t| < R^2$, the cycle of integration is $|y| = r$, $|x| = |t|/r$, $xy = t$, with $|t|/R < r < R$. The integral is independent of r. Thus

$$\frac{1}{2\pi i} \int_{xy=t} f(x,y) \frac{dx \wedge dy}{dt} = \frac{1}{2\pi i} \int f(t/y, y) \frac{dy}{y},$$

and expansion into the series followed by the residue formula leaves only the terms with equal exponents in x and y.

The integration cycle is the Lefschetz vanishing cycle for π near $(0,0)$: if $r = |t|^{1/2}$ the cycle visibly collapses as $|t| \to 0$. This explains the title.

Formula (1) suggests that the operation I has an intrinsic meaning for a morphism of formal schemes $\pi : X \to S$ over k with $X \simeq \mathrm{Spf}\, k[[x,y]]$, $S \simeq \mathrm{Spf}\, k[[t]]$, and π having a non-degenerate quadratic singularity. The two branches of $\pi^{-1}(0)$ are assumed to be defined over k, and an order between them is chosen. Thus we choose coordinates with $\pi(x,y) = xy$, with the first branch of $xy = 0$ being $x = 0$.

Let $\Omega^1_{X/k}$ be the Kähler differentials defined with respect to the adic topology. In coordinates $\Omega^1_{X/k}$ is freely generated by dx, dy, and $\Omega^1_{S/k}$ by dt. Let

$$\omega_{X/k} = \det \Omega^1_{X/k}, \qquad \omega_{S/k} = \Omega^1_{S/k}, \qquad \omega_{X/S} = \omega_{X/k} \otimes \pi^* \omega^{-1}_{S/k}.$$

In coordinates $\omega_{X/S}$ is generated by $(dx \wedge dy)/dt$. The operation I will be interpreted as a morphism

$$\pi_* \omega_{X/S} \longrightarrow \mathcal{O}_S,$$

or, for $S = \mathrm{Spf}(V)$, as a V-linear map

$$H^0(X, \omega_{X/S}) \longrightarrow V.$$

Section 1 provides this interpretation algebraically. The integration sign in (1) is replaced by Grothendieck's local trace. For a uniformizer t of S, the reduction modulo t^n is denoted with the index n. Let A_n be the complete local ring whose formal spectrum is the reduction X_n; in coordinates $A_n = k[[x,y]]/((xy)^n)$. It is convenient to replace X_n by the ordinary scheme $X'_n = \mathrm{Spec}\, A_n$. Let O be its closed point. Grothendieck's local trace is a morphism

$$\mathrm{Tr} : H^1_{\{O\}}(X'_n, \omega) \longrightarrow H^0(S_n, \mathcal{O}).$$

The role of the vanishing cycle is played by an element

$$v_n \in H^1_{\{O\}}(X'_n, \mathcal{O}).$$

In coordinates v_n comes from $\mathrm{Ext}^1_{A_n}(A_n/(x+y)^{2n-1}, A_n)$. In Section 2 these classes are related to an element of the Lie algebra of a Picard scheme. Section 3 globalizes the construction and proves the theorem and the corollary.

1 Algebraic analogue of the integral formula

Keep the notation above. The inclusion $X'_1 \subset X'_n$ is a homeomorphism, since X'_1 is defined by a nilpotent ideal in X'_n. The complement of the origin in X'_n has two connected components corresponding to the branches $x = 0$ and $y = 0$ of X'_1. Let u_n be the section of the structure sheaf of $X'_n - \{O\}$ which is 1 on the first branch and 0 on the second.

Algebraically, the homomorphism

$$A_n = k[[x,y]]/((xy)^n) o k[[x,y]]/(x^n) \times k[[x,y]]/(y^n)$$

induces an isomorphism after inverting $x + y$, and u_n is the element whose image is $(1,0)$. Let v_n be the image of u_n under the boundary map

$$H^0(X'_n - \{O\}, \mathcal{O}) \longrightarrow H^1_{\{O\}}(X'_n, \mathcal{O}).$$

Applying $\mathrm{Ext}^\bullet(-, A_n)$ to

$$0 \to ((x+y)^N) \to A_n \to A_n/((x+y)^N) \to 0$$

gives a diagram showing that, for $N = 2n-1$, v_n lies in the image of

$$\mathrm{Ext}^1_{A_n}(A_n/((x+y)^{2n-1}), A_n).$$

1.2. Proposition. For $f \in k[[x,y]]$ and for every n,

$$\mathrm{Tr}\left(f\frac{dx \wedge dy}{dt}, v_n\right) = -I(f) \pmod{t^n}. \tag{1.2.1}$$

The trace is Grothendieck's local trace. We prove the formula when k is infinite; the general case follows by extension of scalars. Let $I'(f)$ denote the left-hand side of (1.2.1). It is a continuous $k[[t]]/(t^n)$-linear map $A_n \to k[[t]]/(t^n)$. For $\lambda, \mu \in k^*$ let $\sigma(\lambda,\mu)$ be the automorphism

$$(x,y,t) \mapsto (\lambda x, \mu y, \lambda\mu t).$$

It preserves $(dx \wedge dy)/dt$ and v_n, hence

$$I'(\sigma(\lambda, mumu)^* f) = \sigma(\lambda,\mu)^* I'(f). \tag{1.2.2}$$

Applying this to monomials gives $I'(x^a y^b) = 0$ unless $a = b$. By continuity,

$$I'(f) = I(f)I'(1).$$

The scalar $I'(1)$ is independent of n, so it remains to compute it for $n = 1$.

For $n = 1$, $X_1' = \mathrm{Spec}\, k[[x,y]]/(xy)$ is the union of two smooth formal arcs meeting transversely. Sections of ω are pairs of meromorphic differentials (α, β) on the two branches with at most simple poles at O and residues summing to zero. On the punctured curve arbitrary pole orders are allowed. The composite

$$H^0(X_1' - \{O\}, \omega) \to H^1_{\{O\}}(X_1', \omega) \xrightarrow{\mathrm{Tr}} k$$

is the negative of $(\alpha, \beta) \mapsto \mathrm{Res}\,\alpha + \mathrm{Res}\,\beta$. In these terms $(dx \wedge dy)/dt$ and u_1 give residue sum 1, proving (1.2.1).

2 Relation with the Picard functor

Let $S = \mathrm{Spec}\, k[[t]]$, let s be its closed point and η its generic point. Let $\pi : Y \to S$ be proper and flat, purely of relative dimension one, and let $O \in Y_s(k)$. Assume that the completion of Y at O is the formal singularity considered above. Thus Y is regular at O, O is an isolated non-smooth point of π, the singularity is non-degenerate quadratic, and the two branches are defined over k and ordered.

We may further suppose Y regular and connected. Let Y_η be the generic fibre over $k((t))$.

2.2. Lemma. The curve Y_η is absolutely irreducible and generically smooth over $k((t))$.

Indeed, the special fibre is connected and has a rational point, hence is absolutely connected. It is smooth near O away from O, so after a separable extension it has smooth rational points. The corresponding base change has a section; regularity then implies irreducibility, and absolute connectedness gives absolute irreducibility. The smooth locus of the generic fibre is non-empty and therefore dense.

Let C be a complete reduced generically smooth curve over a field K, and D a Cartier divisor on C. The generalized Jacobian $\mathrm{Pic}(C; D)$ represents invertible sheaves on C_T trivialized on D_T. It is smooth, and

$$\mathrm{Lie}\,\mathrm{Pic}(C; D) = H^1(C, \mathcal{O}(-D)). \tag{2.3.1}$$

If C is smooth, the map

$$j : C - D \to \mathrm{Pic}(C; D), \qquad u \mapsto [\mathcal{O}(u(T))] \tag{2.3.2}$$

relates Serre duality with the pullback of invariant differentials on the Picard scheme.

Returning to Y/S, let Z be a relative Cartier divisor disjoint from O, and let $\mathrm{Pic}(Y; Z)^0$ be the relative generalized Picard scheme parametrizing line bundles on Y trivialized along Z and of degree zero on each geometrically irreducible component of each fibre. For each n one constructs morphisms

$$H^0(O_T, \mathcal{O}_T) \longrightarrow H^1(Y_{n,T}, \mathcal{O}(-Z_{n,T})), \tag{2.4.6}$$
$$H^0(O_T, \mathcal{O}_T^*) \longrightarrow H^1(Y_{n,T}, \mathcal{O}(1 \text{ on } Z_{n,T})^*). \tag{2.4.7}$$

The multiplicative version gives a morphism of $k[[t]]/(t^n)$-group schemes

$$w_n : \mathbf{G}_m \longrightarrow \mathrm{Pic}(Y_n, Z)^0. \tag{2.4.8}$$

For $m < n$, the reduction of w_n modulo t^m is w_m; for $n = 1$ this is the morphism associated with the two branches.

Although one cannot take a direct limit to get a morphism $\mathbf{G}_m \to \mathrm{Pic}(Y; Z)^0$, one can restrict to finite subgroup schemes. For every integer m one obtains

$$w : \mu_m \longrightarrow \mathrm{Pic}(Y, Z)^0. \tag{2.4.9}$$

Passing to Lie algebras gives

$$dw : \mathrm{Lie}\, \mathbf{G}_m \longrightarrow \mathrm{Lie}\, \mathrm{Pic}(Y; Z)^0, \tag{2.4.10}$$

and hence an element

$$dw(1) \in H^1(Y, \mathcal{O}(-Z)). \tag{2.4.11}$$

Let $\omega_{Y/S}$ be the relative dualizing sheaf and let φ be a section of $\omega_{Y/S}(Z)$. In formal coordinates near O,

$$\varphi = f(x, y) \frac{dx \wedge dy}{dt}.$$

The restriction of φ to the generic fibre is paired with $H^1(Y_\eta, \mathcal{O}(-Z_\eta))$ by Serre duality.

2.5. Proposition. With the notation above,

$$I(f) = -(dw(1), \varphi). \tag{2.5.1}$$

The right-hand side lies in $k[[t]]$, not merely in $k((t))$. Reducing modulo t^n, the local trace and the global Serre trace are compatible; the proposition is reduced to the following lemma.

2.6. Lemma. The morphism from local cohomology at O to $H^1(Y_n, \mathcal{O}(-Z_n))$ sends v_n to $dw_n(1)$.

The image of v_n is also the image of 1 under (2.4.6). The assertion follows from the description of dw_n via dual numbers over $k[[t]]/(t^n)$ and the natural exact sequence comparing the additive and multiplicative constructions.

3 Proof of the theorem

Let S_0 be a smooth curve over k, let $s \in S_0(k)$, and let S be the completion at s. Let $\pi : Y_0 \to S_0$ be proper and flat, purely of relative dimension one, with Y its base change to S and with a point $O \in Y_s$ such that $(Y/S, O)$ is as above. We suppose, for simplicity, that Y_0 is regular and that π has geometrically connected fibres. Let ω_{Y_0/S_0} be the relative dualizing sheaf.

Let φ be a rational section of ω_{Y_0/S_0}, regular at O. In formal coordinates it can be written

$$\varphi = f(x, y)\frac{dx \wedge dy}{dt}.$$

3.2. Proposition. If k has characteristic $p > 0$, then $I(f)$ is algebraic over the field $k(S_0)$ of rational functions on S_0.

After multiplying φ by a rational function on S_0 vanishing sufficiently at s, and after shrinking S_0, we may assume that for a suitable relative Cartier divisor Z_0 disjoint from O, the form φ is a section of ω on $Y_0 - Z_0$. Let $\mathrm{Pic}(Y_0; Z_0)^0$ be the relative generalized Picard scheme. Via Serre duality, φ corresponds to a section of the dual of the vector bundle

$$\mathrm{Lie}\,\mathrm{Pic}(Y_0; Z_0)^0 = R^1\pi_*\mathcal{O}(-Z_0).$$

After base change to S we have $dw : \mathcal{O} \to \mathrm{Lie}\,\mathrm{Pic}(Y; Z)^0$, and Proposition 2.5 reduces Proposition 3.2 to the following key lemma.

3.3. Key Lemma. If $p > 0$, then dw is algebraic over S_0.

In characteristic p, the inclusion $\mu_p \subset \mathbf{G}_m$ induces an isomorphism on Lie algebras:

$$\mathrm{Lie}(\mu_p) \xrightarrow{\sim} \mathrm{Lie}(\mathbf{G}_m).$$

Modulo t^n, one checks that dw is the differential of the morphism

$$w : \mu_p \to \mathrm{Pic}(Y; Z)^0.$$

It remains to use the rigidity of μ_p: for every flat commutative group scheme P over S_0, the functor $\mathrm{Hom}(\mu_p, P)$ is represented by an etale scheme over S_0. Applying this to $P = \mathrm{Pic}(Y_0; Z_0)^0$, the morphism w is pulled back from a universal morphism over an etale cover of S_0. Hence its differential, and therefore dw, is algebraic.

Let $k\{x_1, \ldots, x_N\}$ be the henselization of the local ring of affine N-space at the origin. Its completion is $k[[x_1, \ldots, x_N]]$, and an element of the completion is algebraic over $k(x_1, \ldots, x_N)$ if and only if it lies in the henselization.

Proof of the theorem

If f is algebraic, there exists a surface Y_0 with a rational point O, an etale morphism $h : Y_0 \to \mathbf{A}^2$ sending O to the origin, and a section whose formal expansion at O is f. Let $S_0 = \mathbf{A}^1$ with coordinate t, and let π' be the composite

$$Y_0 \xrightarrow{h} \mathbf{A}^2 \xrightarrow{xy} \mathbf{A}^1 = S_0.$$

The differential form $f\,h^*((dx \wedge dy)/dt)$ is a rational section of the relative dualizing sheaf. After compactifying and resolving singularities, one obtains a proper flat regular model to which Proposition 3.2 applies. Therefore $I(f)$ is algebraic.

Proof of the corollary

Proceed by induction on N. The cases $N = 0$ and $N = 1$ are immediate, and $N = 2$ is the theorem. For $N > 2$, write

$$g = \sum g_{n,m} x_{N-1}^n x_N^m, \qquad g_{n,m} \in k[[x_1, \dots, x_{N-2}]],$$

and put

$$I(g) = \sum g_{n,n} x_{N-1}^n.$$

Then

$$I_N(g) = I_{N-1}(I(g)). \tag{3.6.1}$$

If g is algebraic, the coefficients $g_{n,m}$ are algebraic over $k(x_1, \dots, x_{N-2})$. Applying the theorem over the fraction field of the henselization in the first $N - 2$ variables shows that $I(g)$ is algebraic, and induction proves the claim.

Degree estimates and remarks

The proof gives an estimate. With the notation of the geometric construction, let α be the degree of the curve S_0 over the t-line, let g be the genus of the normalization of the geometric generic fibre, and let $z = \max(0, z' - 1)$ where z' is the number of geometric points of the divisor Z. Then the degree d of the equation satisfied by $I(f)$ is bounded by

$$d < \alpha p^{g+z}. \tag{3.7.1}$$

For a formal algebraic series with integer coefficients, this implies that for almost all primes p, the degree d_p of the reduction of $I(f)$ modulo p satisfies

$$d_p < \alpha p^{g+z}. \tag{3.7.2}$$

Using μ_{p^m} when $p^m = 0$ on the base gives analogous information modulo powers of p, for almost all primes.

A direct proof for N variables would be preferable. Analytically, for a convergent series g one expects the formula

$$I(g)(t) = \int_{Z(t)} g \, \frac{dz_1 \cdots dz_N}{dt}, \tag{1'}$$

where

$$Z(t) = \{(x_1, \dots, x_N) : x_1 \cdots x_N = t, |x_i| = r_i, \prod r_i = |t|\}.$$

A direct proof of this kind would be a test case for a p-adic theory of vanishing cycles.

Furstenberg's proof in the rational case shows that if g is one component of a column vector h with entries in $k[[x_1, \dots, x_N]]$ satisfying

$$h = Ah^p$$

for a square matrix A with polynomial entries, then $I(g)$ has the same property and is algebraic. This type of equation recalls F-crystals in crystalline cohomology. Since $T_p \operatorname{Pic}$ is an avatar of p-adic H^1, the proof here and Furstenberg's proof are perhaps less distant from each other than they first appear.

A Signs

The treatment of signs in the duality formalism for coherent algebraic sheaves is delicate. We record the conventions used here. For a flat relative complete intersection morphism $f : X \to S$, purely of relative dimension n, the relative dualizing sheaf $\omega_{X/S}$ is related to the relative dualizing complex by

$$K_{X/S} = \omega_{X/S}[n]. \tag{S.1}$$

When X is smooth over S, $\omega_{X/S} = \Omega^n_{X/S}$. If X is the fibre product over S of morphisms $f_i : X_i \to S$ of relative dimensions n_i, then

$$K_{X/S} = \boxtimes_i K_{X_i/S}. \tag{S.2}$$

Equivalently, after choosing a total order on the index set,

$$\omega_{X/S} = \bigotimes_i \omega_{X_i/S}, \tag{S.3}$$

with the sign change $(-1)^N$ when the order is changed, where N is the sum of $n_i n_j$ over inverted pairs.

For $f : X \to S$ proper as above, the trace is

$$\mathrm{Tr} : Rf_* K_{X/S} \to \mathcal{O}_S. \tag{S.4}$$

It induces

$$\mathrm{Tr} : R^n f_* \omega_{X/S} \to \mathcal{O}_S. \tag{S.5}$$

For a section s of f there is a local trace

$$\mathrm{Tr} : Rs^! K_{X/S} \to \mathcal{O}_S. \tag{S.6}$$

The choices are fixed by requiring compatibility between global and local traces and by normalizing, for a smooth curve X over an algebraically closed field and a point $x \in X(k)$, the local trace

$$H^1_{\{x\}}(X, \Omega^1[1]) \to k \tag{S.7}$$

to be the residue map Res_x.

With these conventions, for a smooth complete curve X and $x \in X(k)$, the boundary map

$$H^0(X - \{x\}, \Omega^1) \to H^1_{\{x\}}(X, \Omega^1)$$

composed with the trace is $-\mathrm{Res}_x$. For products, if $X = X_1 \times X_2$ and $u_i \in H^{n_i}(X_i, \omega_{X_i})$, then

$$\mathrm{Tr}(\mathrm{pr}_1^* u_1 \smile \mathrm{pr}_2^* u_2) = (-1)^{n_1 n_2} \mathrm{Tr}(u_1) \mathrm{Tr}(u_2).$$

Over $k = \mathbf{C}$, comparison with the transcendental trace identifies the algebraic trace with integration over $X(\mathbf{C})$, with the standard normalization by $(2\pi i)^{-n}$. For projective space $\mathbf{P}^n$ with homogeneous coordinates $T_0, \ldots, T_n$ and $t_i = T_i/T_0$, the Cech class of

$$\frac{dt_1 \wedge \cdots \wedge dt_n}{t_1 \cdots t_n}$$

has trace $(-1)^{n(n+1)/2}$ with these conventions.

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